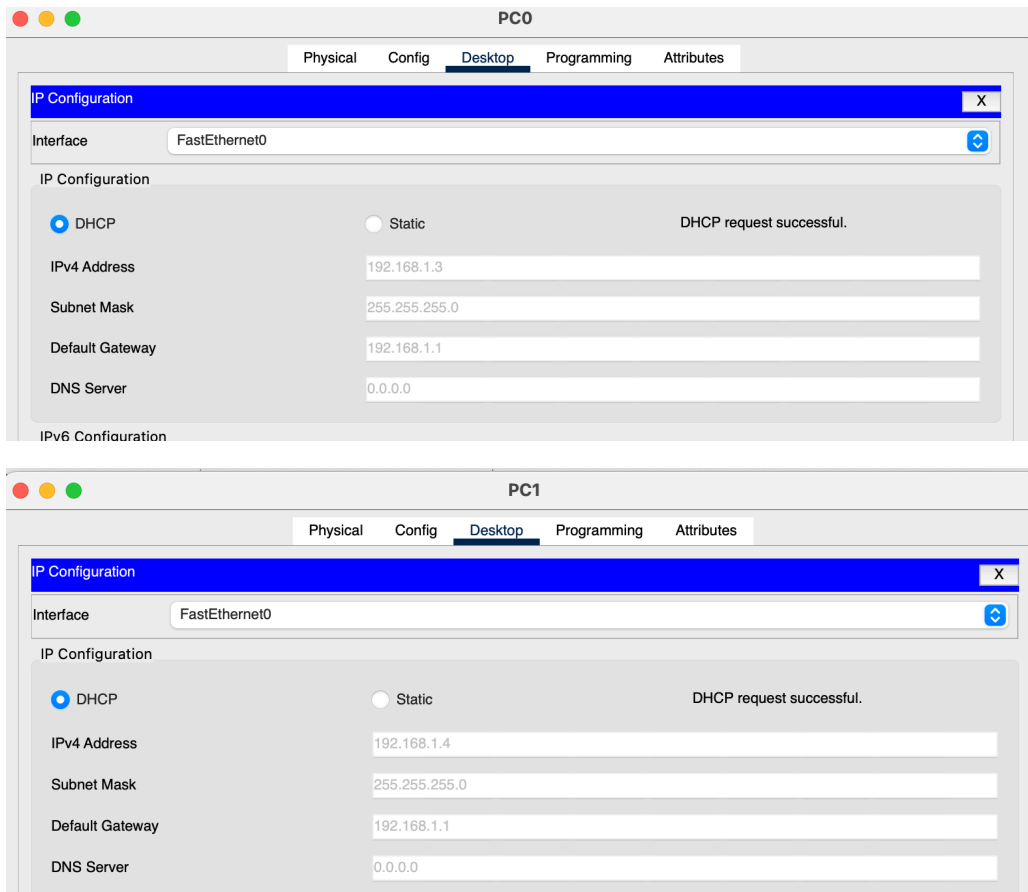


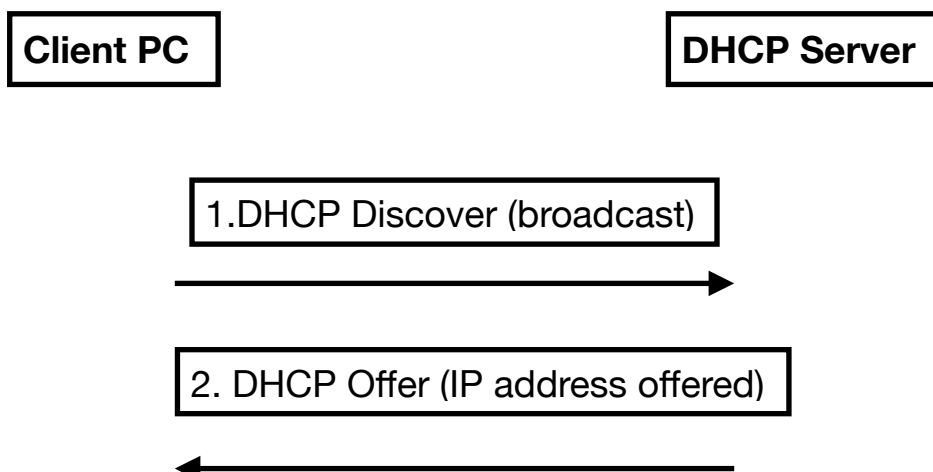
Module 2: Network + Network Security , Maintenance and Troubleshooting

Day 2 DHCP Configuration

Task 1:



Task 2:



2. DHCP Request (client accepts offer)



2. DHCP Acknowledge (lease confirmed)



Task 3:

PC 1

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

PC 2

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

DHCP request failed.

IPv4 Address:

Subnet Mask:

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Since the DHCP pool was restricted to Max Users = 1, only one IP address (192.168.1.3) is available for lease. When PC1's DHCP Discover message is processed first, the server allocates that single address and marks it as leased. When PC2 sends its own Discover, the server has no addresses left in the pool to offer, so PC2 fails to receive an IP configuration.

Task 4:

In a real café like Zomato's partner restaurants offering customer WiFi, dozens of devices connect and disconnect constantly throughout the day. DHCP eliminates the need for staff to manually configure an IP address for every customer's phone or laptop — it automatically assigns a unique, non-conflicting address to each device the moment it joins the network, and reclaims that address when the device leaves (lease expiry), making it efficient and scalable for public WiFi with high device turnover.